🌐 Module 13: IoT Penetration Testing

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# Exercise 04: Probing IOT Devices by Analyzing UPnP and mDNS Messages
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Scenario

Zero Configuration Networking (Zeroconf) Security & mDNS Spoofing analyzes automated address assignment, service discovery, and hostname resolution protocols—specifically **Multicast DNS (mDNS)** operating over UDP port `5353` and **Universal Plug and Play (UPnP)**.

Key technical concepts in mDNS and UPnP IoT probing:
1. **mDNS Probing & Announcement Phase**: When an mDNS responder initializes or changes network state, it broadcasts `ANY` resource queries over UDP multicast (`224.0.0.251:5353`).
2. **mDNS Multicast Spoofing & Name Poisoning**: Because mDNS responses are unauthenticated multicast traffic, tools like **Pholus** (**pholus3.py**) continuously send spoofed mDNS responses. This causes target IoT nodes (such as Ubuntu booting `_ipp._tcp.local`) to continuously collision-increment hostnames (`ubuntu-2.local`, `ubuntu-3.local`) and fail local registration.
3. **Pholus Reconnaissance Modes**:
- Query passive discovery mode: **sudo python pholus3.py eth0 -rq -stimeout 10**
- Active spoofing & response enumeration mode: **sudo python pholus3.py eth0 -afre -stimeout 1000**

In this exercise, we log into **Parrot Security** (**pentester** / **toor**), navigate to **~/Pholus**, capture background mDNS packets in Wireshark on interface `eth0`, execute Pholus passive query probes, launch active response enumeration (`-afre`), capture mDNS broadcasts from target Ubuntu machine (**192.168.0.19**), and record host IP configuration.

Objective

The objective of this lab is to help students learn how to:
- Authenticate to Parrot Security target VM (**pentester** / **toor**)
- Navigate to Pholus framework directory: **cd ~/Pholus**
- Launch Wireshark to monitor UDP port `5353` multicast traffic on `eth0`
- Execute passive mDNS query probe: **sudo python pholus3.py eth0 -rq -stimeout 10**
- Execute active mDNS response and service discovery enumeration: **sudo python pholus3.py eth0 -afre -stimeout 1000**
- Disable secondary interface if required: **sudo ifconfig eth1 down**
- Capture mDNS boot-up announcements (`_ipp._tcp.local`) and identify target IP (**192.168.0.19**)

Task 1: Execute Pholus mDNS Probing, Analyze Zeroconf Messages, and Identify Ubuntu Host IP

In this task, we launch Wireshark on `eth0`, execute Pholus mDNS passive and active probing scripts, capture Zeroconf service announcements, and record host IP `192.168.0.19`.

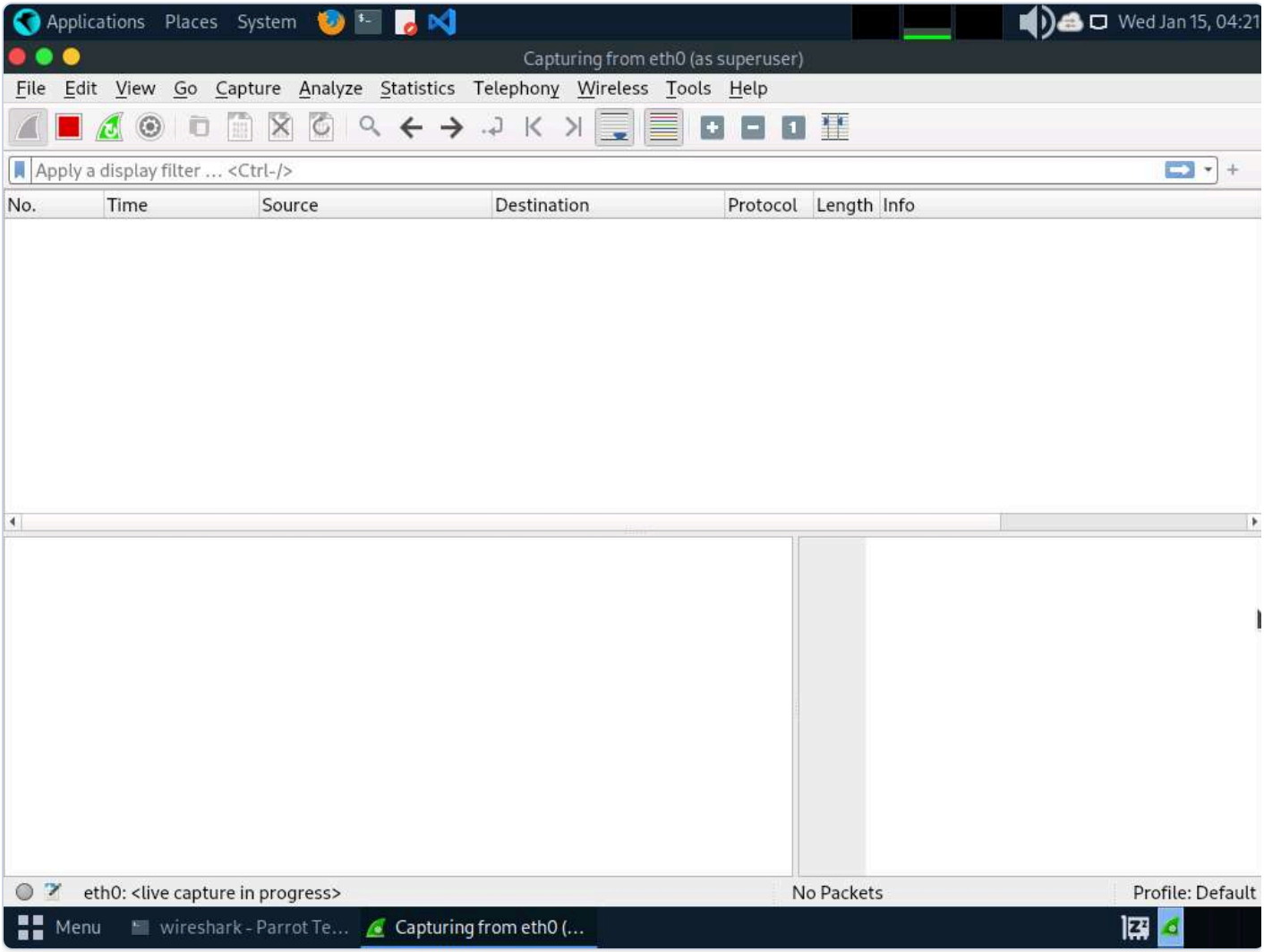
Lab Tasks

1. Log into **Parrot Security** VM using credentials (**pentester** / **toor**). Open terminal and navigate to Pholus directory: **cd ~/Pholus**

2. Review mDNS UDP port `5353` multicast query/response flow architecture.

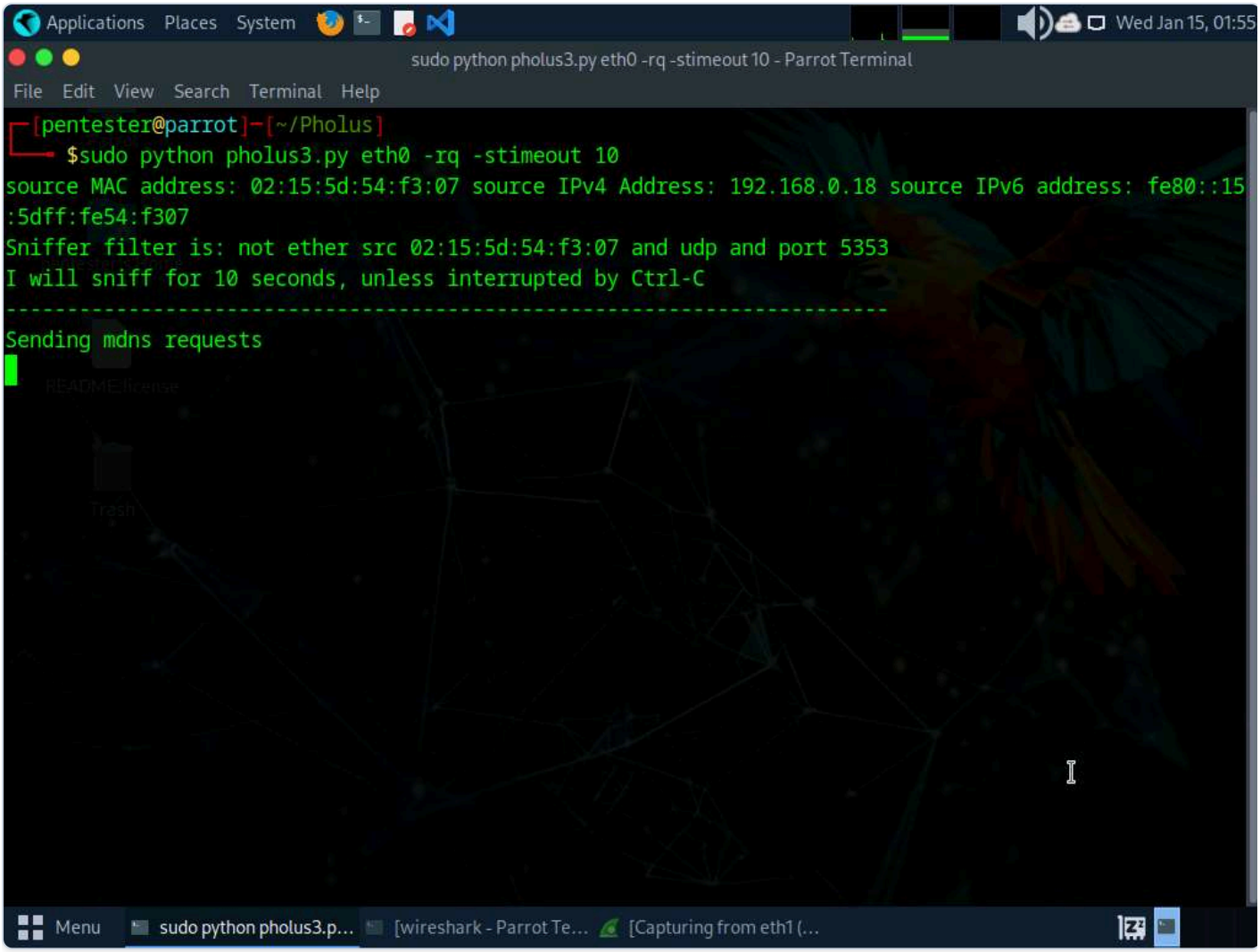
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▸ Frame 1075: 88 bytes on wire (704 bits), 88 bytes captured (704 bits) on interface ens36, id 0
▸ Ethernet II, Src: VMware_22:ab:f9 (00:0c:29:22:ab:f9), Dst: IPv4mcast_fb (01:00:5e:00:00:fb)
▸ Internet Protocol Version 4, Src: 192.168.100.132, Dst: 224.0.0.251
▸ User Datagram Protocol, Src Port: 5353, Dst Port: 5353
▾ Multicast Domain Name System (query)
    Transaction ID: 0x0000
  ▸ Flags: 0x0100 Standard query
    Questions: 1
    Answer RRs: 0
    Authority RRs: 0
    Additional RRs: 0
  ▾ Queries
    ▾ _services._dns-sd._udp.local: type ANY class IN, "QM" question
        Name: _services._dns-sd._udp.local
        [Name Length: 28]
        [Label Count: 4]
        Type: * (A request for all records the server/cache has available) (255)
        .000 0000 0000 0001 = Class: IN (0x0001)
        0... .... .... .... = "QU" question: False
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1. Open Wireshark with `sudo` privileges and start packet capture on interface `eth0`.

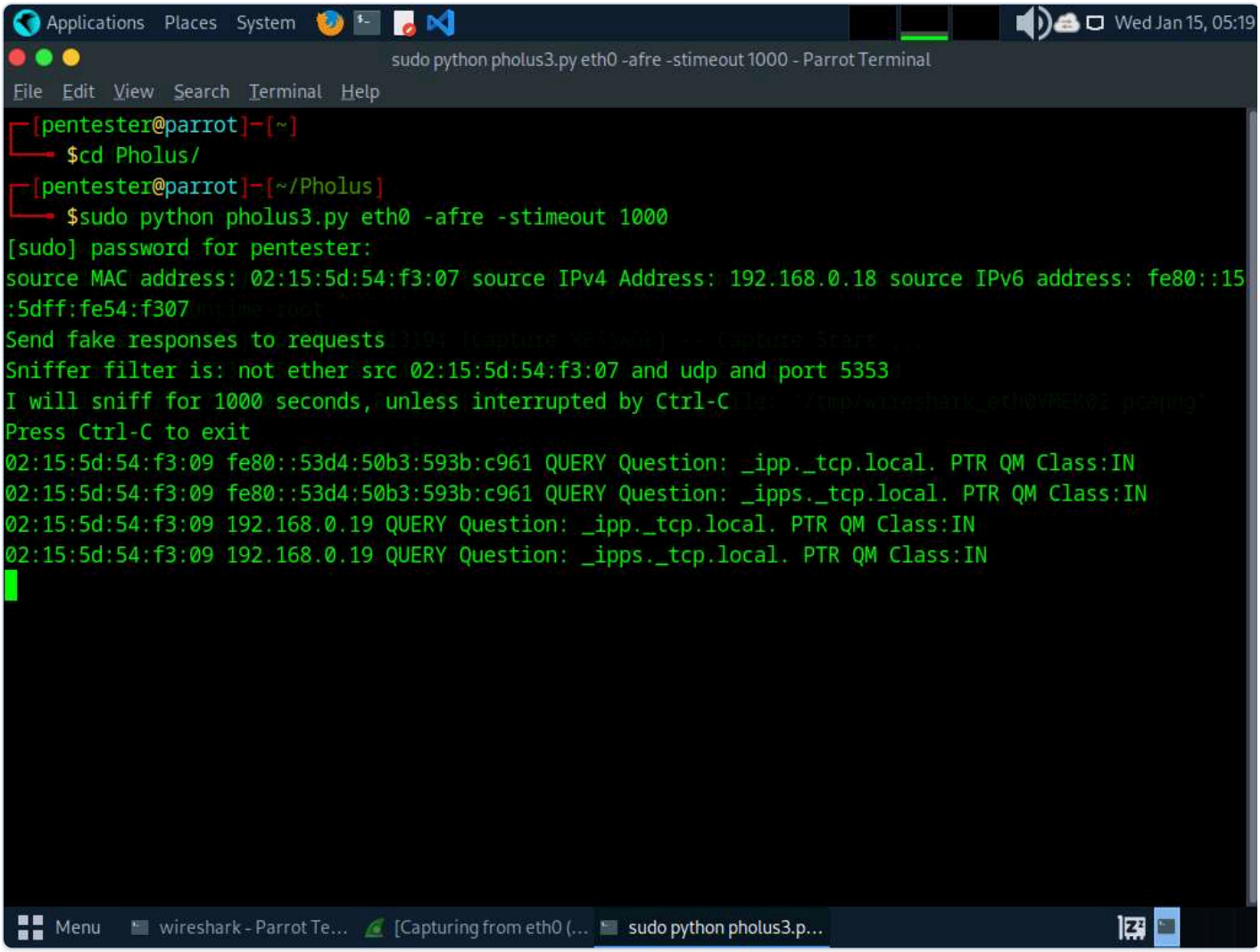


1. Execute passive mDNS query probing script:

   sudo python pholus3.py eth0 -rq -stimeout 10



1. Execute active mDNS service enumeration and name poisoning mode (bring secondary interface down if necessary with `sudo ifconfig eth1 down`):
   sudo python pholus3.py eth0 -afre -stimeout 1000
   Observe output: Captured boot-up mDNS announcements for `_ipp._tcp.local` from target Ubuntu host IP **192.168.0.19**.



This concludes the lab exercise.

> **IMPORTANT:** ❓ **Question 13.4.1.1** On the Parrot Security machine, use Pholus send the mDNS queries to capture the replies from Ubuntu machine. Enter the IP Address of the Ubuntu machine whose boot up name is captured. [!SUCCESS] ✅ **Answer:** **192.168.0.19**